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CS6903: Network Security

(Introduction to Libpcap)

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Acknowledgement

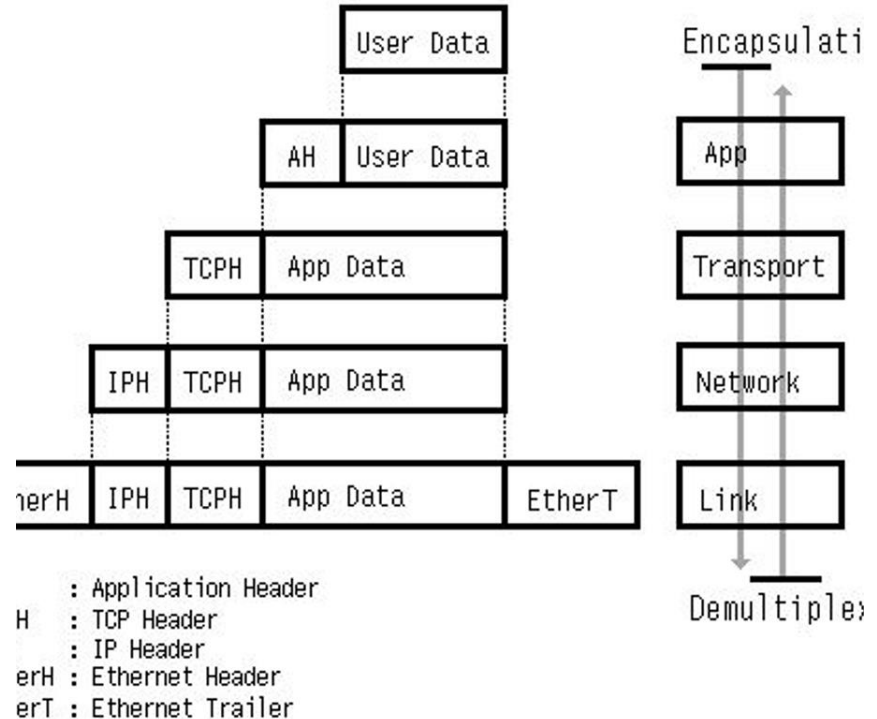
- ❑ POSTECH DP&NM Lab
(<http://dpmn.postech.ac.kr/cs353/2019/src/Programming-using-Libpcap.pdf>)
- ❑ <https://yuba.stanford.edu/~casado/pcap/section1.html>
- ❑ Various other sources on Internet

Introduction

- ❑ Libpcap: Portable Packet Capturing Library
 - OS independent
 - Simple and powerful user-level library
 - Compatible with Unix like system
 - Many of commercial IDS systems utilizes Libpcap to analyze packet data
- ❑ Representative Programs Rely on Libpcap
 - TCPDump, ntop, ZEEK, snort, Wireshark, ethereal, etc...
- ❑ Other packet capturing tools
 - SOCK_PACKET, LSP, SNOOP, SINT, etc...
 - OS Dependent
- ❑ Requires **super user (root)** privileges when capturing live data

Basic Concept of Packet Capturing

- ❑ Packet capturing does not affect data transfer
- ❑ The packet capture by libpcap is called raw packet, and demultiplexing is required to analyse the packet



Libpcap File Format

- Normally has “.pcap” file extension

- General libpcap file format

- Contains some global information followed by zero or more records for each packet



- A captured packet in a capture file does not necessarily contain all the data
- A captured file might contain at most first N bytes of each packet

- Global Header

```
typedef struct pcap_hdr_s {  
    guint32 magic_number;    /* magic number */  
    guint16 version_major;   /* major version number */  
    guint16 version_minor;   /* minor version number */  
    gint32  thiszone;        /* GMT to local correction */  
    guint32 sigfigs;         /* accuracy of timestamps */  
    guint32 snaplen;         /* max length of captured packets, in octets */  
    guint32 network;         /* data link type */  
} pcap_hdr_t;
```

Device & Network related APIs (1/2)

❑ Device & Network lookup for single device

➤ `char *pcap_lookupdev(char *errbuf)`

- Return a pointer to a network device suitable for use with `pcap_open_live()` and `pcap_lookupnet()`
- Return NULL indicates an error
- Reference: `lookupdev.c`

➤ `int pcap_lookupnet(const char *device, bpf_u_int32 *net0, bpf_u_int32 *maskp, char *errbuf)`

- Determine the network number and mask associated with the network device
- Return -1 indicated an error
- Reference: `lookupnet.c`

Device & Network related APIs (2/2)

□ Device & Network lookup for multiple device

- `int pcap_findalldevs(pcap_if_t **alldevsp, char *errbuf)`
 - Constructs a list of network devices that can be opened with `pcap_create()` and `pcap_activate()` or with `pcap_open_live()`
 - `alldevsp`: list of network devices
 - Returns 0 on success and -1 on failure
 - List of devices must be freed with `pcap_freealldevs()`
- Structure of `pcap_if_t`
 - `next`: if not NULL, a pointer to the next element in the list
 - `name`: a pointer to a string giving a name for the device to pass to `pcap_open_live()`
 - `description`: if not NULL, a pointer to a string giving human-readable description of the device
 - `addresses`: a pointer to the first element of a list of addresses
 - `flags`: interface flags – `PCAP_IF_LOOPBACK` set if the interface is a loopback interface

Initializing Packet Capturing APIs

❑ Preparation of Packet Capturing

- `pcap_t *pcap_open_live(const char *device, int snaplen, int promisc, int to_ms, char *errbuf)`
 - Parameters
 - `device`: the device in which the packets are captured from
 - `snaplen`: maximum number of bytes to capture
 - `promisc`: true. Set the interface into promiscuous mode; false, only bring packet intended for you
 - `to_ms`: read timeout in milliseconds, zero, cause a read to wait forever to allow enough packets to arrive
 - Return
 - A packet capture descriptor to look at the packets on the network
 - Return NULL indicates an error

Initializing Packet Capturing APIs...

□ Preparation of Packet Capturing

- `pcap_t *pcap_open_offline(const char *fname, char *errbuf)`
 - Open a saved file for reading
 - `fname`: the name of the file to open
 - Return a `pcap_t *` on success and `NULL` on failure

Packet Read Related APIS

❑ `const u_char *pcap_next(pcap_t *p, struct pcap_pkthdr *h)`

- Read the next packet
- Return NULL indicated an error
- Reference: `pcap_next.c`

❑ `int pcap_loop(pcap_t *p, int cnt, pcap_handler callback, u_char *user)`

- Process packets from a live capture of “saved file” until cnt packets are processed
- A value of -1 or 0 for cnt is equivalent to infinity
- Callback specifies a routine to be called

Compile Libpcap Program

❑ Install libpcap

- `sudo apt install libpcap-dev`

❑ Compile

- `gcc -o test test.c -lpcap`